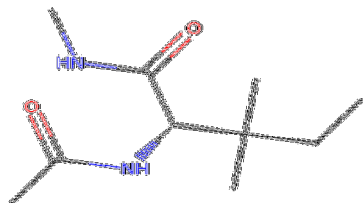


Semantic Decomposition of NCAA Building Blocks

Chemical Structure

Semantic Token Properties

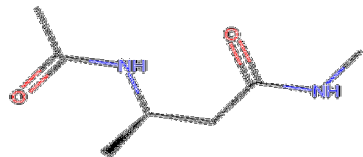


α-amino acid (isoleucine analog)



[A01]

class	I
chirality	L
charge	neu
backbone	alpha
N-mod	NO
bulk	large
polarity	low
flexibility	med

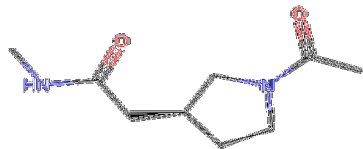


β-amino acid (β-alanine analog)



[N15]

class	A
chirality	D
charge	neu
backbone	beta
N-mod	NO
bulk	small
polarity	low
flexibility	med



γ-amino acid (γ-proline analog)



[s01]

class	P
chirality	L
charge	neu
backbone	gamma
N-mod	NCY
bulk	small
polarity	low
flexibility	low